Answer Key - Biology - Undergraduate Level Assessment 6 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) growth hormone; B) estrogen; C) thyroxine; D) insulin; E) prolactin
   Note: Thyroxine is the hormone that regulates metabolic processes and stimulates the developmental changes necessary for metamorphosis in amphibians, particularly the transformation from tadpole to adult form.

2. **Answer:** A
   Options: A) The number of synapses crossed; B) The speed of the action potential; C) The amplitude of the action potential; D) The number of neurons activated; E) The refractory period of the neuron
   Note: The correct answer is based on the principle that the greater the intensity of a stimulus, the more sensory neurons are activated and synapses crossed, leading to an increased signal that allows the brain to detect differences in stimulus intensity.

3. **Answer:** C
   Options: A) 50%; B) 100%; C) 0%; D) 10%; E) 75%
   Note: The child cannot have type O blood because type O blood requires two O alleles, while the parent with type AB blood has no O alleles to contribute.

4. **Answer:** A
   Options: A) are more abundant than most other species in their communities; B) tend to reduce diversity by eliminating food resources for other species; C) are the largest species in their communities; D) are the most aggressive species in their communities; E) are the only species capable of reproduction
   Note: correct because keystone species, despite not necessarily being the most abundant, play a critical role in maintaining the structure and diversity of their ecological communities through their unique interactions and effects on other species.

5. **Answer:** A
   Options: A) Pancreas; B) Thymus; C) pituitary gland; D) adrenal gland; E) thyroid gland
   Note: A deficiency in the pancreas can lead to imbalances in calcium metabolism due to disrupted insulin and glucagon secretion, which can ultimately result in tetany.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: This statement is false because different tissue types in higher plants, such as

dermal, vascular, and ground tissues, have distinct structural and functional roles that contribute to the overall complexity of plant organization and classification.

7. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because organ transplants are generally successful when performed, although the challenge lies in the limited availability of donor organs, not the inherent success rate of the transplants themselves.

8. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because acid rain introduces additional hydrogen ions (H+) into the lakes, thereby lowering the pH and indicating increased acidity.

9. **Answer:** False
   Options: A) True; B) False
   Note: A single nucleotide insertion disrupts the reading frame of the mRNA, leading to a completely different amino acid sequence downstream, whereas a deletion of a nucleotide triplet typically removes one amino acid without affecting the overall reading frame.

10. **Answer:** False
    Options: A) True; B) False
    Note: Mules are hybrids resulting from the mating of a horse and a donkey, and they are typically sterile, meaning they cannot reproduce either sexually or asexually to pass on genetic traits.

Long Form Response

11. **Answer:** The removal of product C from the equilibrium reaction A + B $\rightarrow$ C, catalyzed by enzyme K, can significantly influence the production of more product C due to Le Chatelier's principle. According to this principle, if a system at equilibrium experiences a change in concentration, temperature, or pressure, the system will adjust to counteract that change and restore a new equilibrium. By removing product C, the concentration of C decreases, shifting the equilibrium to the right to favor the formation of more C from A and B. This shift results in an increased production of product C, demonstrating how manipulating reactant or product concentrations can affect biochemical reactions. Thus, the removal of C effectively drives the reaction forward, enhancing the overall yield of the desired product.
    Note: correct because it accurately applies Le Chatelier's principle, stating that the removal of product C decreases its concentration, prompting the equilibrium to shift right to increase the production of C from reactants A and B.

12. **Answer:** A cross between two brown-furred minks, both with the genotype Bb (where 'B' represents the dominant allele for brown fur and 'b' represents the recessive allele for silver fur), can produce silver-furred offspring due to the principles of Mendelian

inheritance. In this scenario, the offspring can inherit combinations of alleles from each parent: BB, Bb, and bb. The genotype bb results in silver fur, as it expresses the recessive trait. When two Bb minks are crossed, there is a 25% chance that the offspring will be homozygous recessive (bb), thus resulting in silver-furred minks. This demonstrates how dominant and recessive alleles interact in inheritance patterns, leading to phenotypic variation within the offspring.
Note: correct because it accurately describes the genetic cross between two heterozygous brown-furred minks, illustrating how the inheritance of alleles can lead to the expression of the recessive silver fur trait in 25% of the offspring.

End of Answer Key